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ICSE CLASS 10 · 2027 SYLLABUS

Mathematics Cheat Sheet

Every formula, identity and theorem from ICSE Class 10 Mathematics. 8 pages. Nothing missing.

GST & Banking | Shares | Quadratics | Factor Theorem | Matrices | AP & GP | Coordinate Geometry | Similarity | Circles | Mensuration | Trigonometry | Statistics | Probability

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01 Commercial mathematics

Profit, loss & discount

Profit	SP − CP (if SP > CP)
Loss	CP − SP (if CP > SP)
Profit %	$\frac{\text{Profit}}{\text{CP}} \times 100$
Loss %	$\frac{\text{Loss}}{\text{CP}} \times 100$
Discount	List price × discount %
Price after discount	List price − Discount
SP from profit %	$\text{CP} \times \frac{100 + \text{P}}{100}$
CP from SP (profit)	$\text{SP} \times \frac{100}{100 + \text{P}}$

GST: Goods & Services Tax

- **Taxable value** = List price − Discount (tax is charged on the discounted price).

GST amount	Taxable value × $\frac{\text{rate}}{100}$
Bill amount	Taxable value + GST
Intra-state sale	GST = CGST + SGST
CGST and SGST	each = ½ of GST
Inter-state sale	IGST (full rate)

- **Inverse case:** price including GST = M × (1 + r/100). So M = $\frac{\text{Price paid} \times 100}{100 + \text{r}}$.

Recurring deposit (RD)

- **P** = monthly instalment, **n** = number of months, **r** = rate % p.a.

$$I = P \times \frac{n(n+1)}{2} \times \frac{r}{12 \times 100}$$

$$MV = P \times n + I$$

Total deposit	P × n
Interest only	MV − P × n

- **Time in years:** n/12. Each month the instalment earns interest for different periods: first for n months, last for 1 month.

Shares & dividends

Term	Meaning
Face (nominal) value, FV	Value printed on the share
Market value, MV	Price at which it is bought or sold
Premium / above par	MV > FV (e.g., ₹100 share at ₹120)
Discount / below par	MV < FV
At par	MV = FV

Number of shares	Investment ÷ MV
Investment	N × MV
Dividend (income)	N × rate × FV
Rate of dividend	(dividend per share ÷ FV) × 100
Return %	(Income ÷ Investment) × 100

02 Inequations & quadratics

Linear inequations

- Solve like an equation with ONE exception:

Multiply or divide by a negative → reverse the sign

Example: $-3x > 9 \rightarrow x < -3$

- **Set notation:** $\{x : x \in R, x < -3\}$.

Replacement set	Meaning
N	Natural: 1, 2, 3, ...
W	Whole: 0, 1, 2, ...
Z	Integers: ..., −1, 0, 1, ...
R	All real numbers

- **Number line:** a **hollow** circle for < or >. A **solid** circle for ≤ or ≥. Shade the solution direction.

Proportion

a : b = c : d	$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$

- **Extremes:** a, d. **Means:** b, c. Product of extremes = product of means.

Continued proportion	$a : b = b : c \Rightarrow b^2 = ac$
Mean proportion of a, c	$\sqrt{ac}$
Third proportional	$c = b^2/a$
Fourth proportional	$d = bc/a$

Quadratic equation

$$ax^2 + bx + c = 0, a \neq 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant	$D = b^2 - 4ac$
Sum of roots	$-b / a$
Product of roots	c / a

Condition	Nature of roots
$D > 0$	Two **distinct real** roots
$D = 0$	Two **equal real** roots ($x = -b/2a$)
$D < 0$	**No real** roots

03 Ratio, proportion & factor theorem

Properties of proportion

Property	If a/b = c/d, then
Invertendo	b/a = d/c
Alternendo	a/c = b/d
Componendo	(a + b)/b = (c + d)/d
Dividendo	(a − b)/b = (c − d)/d
Componendo-dividendo	(a + b)/(a − b) = (c + d)/(c − d)

Remainder theorem

$$f(x) \div (x - a) \rightarrow \text{remainder} = f(a)$$

Divisor (ax + b) — **remainder = f(−b/a)**

Example: $f(x) = x^3 - 2x + 5$ divided by $(x - 2)$: $f(2) = 8 - 4 + 5 = \mathbf{9}$.

Factor theorem

$$(x - a) \text{ is a factor of } f(x) \iff f(a) = 0$$

- **To factorise a cubic:** (1) try $x = \pm1, \pm2, \pm3$ (factors of the constant term) until $f(a) = 0$. (2) Divide $f(x)$ by $(x - a)$. (3) Factorise the quadratic.
- **Degree** of $f(x)$ does not exceed 3 in this syllabus.

04 Matrices, AP & GP

Matrix basics

- **Order** = rows × columns. A matrix with m rows and n columns is m × n.

Type	Description
Row matrix	1 × n
Column matrix	m × 1
Square	m = n
Null (zero)	all elements 0
Identity I	diagonal 1s, rest 0

I (2×2) — $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Equality — **same order and equal elements**

Operations

- **Add / subtract:** only if **same order**; add or subtract matching elements.
- **Scalar × matrix:** multiply every element by k.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \times \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$$

- **Multiplication is possible** only if columns of A = rows of B. If A is m × n and B is n × p, then AB is m × p.
- **AB ≠ BA** in general. **A(B + C) = AB + AC.**

Arithmetic progression (AP)

General term	$T_n = a + (n - 1)d$
Common difference	$d = T_2 - T_1$
Sum of n terms	$S_n = \frac{n}{2}[2a + (n - 1)d]$
Sum (with last l)	$S_n = \frac{n}{2}(a + l)$
T_n from sums	$T_n = S_n - S_{n-1}$
Three terms in AP	$a - d, a, a + d$
b is AP mean of a, c	$2b = a + c$
Sum 1 + 2 + … + n	$n(n + 1)/2$

Geometric progression (GP)

General term	$T_n = a\,r^{n-1}$
Common ratio	$r = T_2 / T_1$
Sum, r > 1	$S_n = \frac{a(r^n - 1)}{r - 1}$
Sum, r < 1	$S_n = \frac{a(1 - r^n)}{1 - r}$
Three terms in GP	a/r, a, ar
b is GP mean of a, c	$b^2 = ac$

05 Coordinate geometry

Reflection of a point (x, y)

In	Image
x-axis (y = 0)	(x, −y)
y-axis (x = 0)	(−x, y)
Origin	(−x, −y)
Line x = a	(2a − x, y)
Line y = b	(x, 2b − y)

- **Invariant point:** a point that does not move after reflection. Any point **on the mirror line** is invariant. For reflection in the origin, only (0, 0).

Section & midpoint formula

- **Internal division** of AB in the ratio $m_1 : m_2$:

$$P = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$$

Midpoint	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
Centroid	$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$
Distance	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Similar triangles

- **Similar:** same shape, different size. Corresponding angles equal and sides proportional.

Test	Condition
AA	Two angles equal
SSS	Three sides proportional
SAS	One angle equal; sides including it proportional

- **Congruent:** same shape and size. Similar triangles have the **same shape only**.

Slope (gradient)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan\theta$$

- **θ** is the angle the line makes with the positive x-axis (anticlockwise).

Line	Slope
Rises left to right	Positive
Falls left to right	Negative
Horizontal (∥ x-axis)	0
Vertical (∥ y-axis)	Not defined

Parallel lines	$m_1 = m_2$
Perpendicular lines	$m_1 \times m_2 = -1$
Collinear A, B, C	slope AB = slope BC

Equation of a line

$$y = mx + c$$

m	slope
c	y-intercept (line cuts y-axis at (0, c))
Point-slope	$y - y_1 = m(x - x_1)$
Two-point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
x-axis	y = 0
y-axis	x = 0
Parallel to x-axis	y = k
Parallel to y-axis	x = h
x-intercept	put y = 0

06 Similarity & loci

Basic proportionality theorem (BPT)

- **If DE ∥ BC** in △ABC (D on AB, E on AC):

$$AD / DB = AE / EC$$

Also	AD / AB = AE / AC

- **Converse:** if AD/DB = AE/EC then DE ∥ BC.
- **Midpoint theorem:** the line joining midpoints of two sides is ∥ to the third and half of it.

Areas & scale factor

$$\frac{\text{Area } \triangle ABC}{\text{Area } \triangle PQR} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{h_1}{h_2}\right)^2$$

If lengths scale by k	Then
Length	k
Area	k^2
Volume	k^3

- **Maps and models:** scale 1 : n means actual length = n × map length; actual area = n^2 × map area.

Same height — **areas ∝ bases**

Locus

- **Locus:** path of a point that moves under a given condition.

Condition	Locus
Fixed distance r from a fixed point O	A **circle**, centre O, radius r
Equidistant from two intersecting lines	**Angle bisectors** of the angles between them
Equidistant from two points A and B	**Perpendicular bisector** of AB
Equidistant from two parallel lines	A line parallel and midway
Fixed distance from a line	Two parallel lines on either side

07 Circles

Angle properties

- **1.** The angle an arc subtends at the **centre** = **2×** the angle at any point on the remaining circle.

$$\angle AOB = 2\,\angle ACB$$

- **2.** Angles in the **same segment** are equal.
- **3.** Angle in a **semicircle** = 90°.
- **4.** Equal chords subtend equal angles at the centre.
- **5.** A perpendicular from the centre to a chord **bisects** the chord.

Cyclic quadrilateral

- **1.** Opposite angles are **supplementary**: ∠A + ∠C = 180° and ∠B + ∠D = 180°.
- **2. Exterior angle** = interior **opposite** angle.
- **Converse:** if opposite angles are supplementary, the quadrilateral is cyclic.

Tangent properties

- **1.** Tangent ⊥ radius at the point of contact.
- **2.** Two tangents from an external point are **equal in length**: PA = PB.
- **3.** If two circles touch, the point of contact lies on the **line joining their centres**.

Touching externally — $d = r_1 + r_2$

Touching internally — $d = r_1 - r_2$

Chords & secants: the three lengths rules

Intersecting chords: $PA \times PB = PC \times PD$

- **Internal or external:** works for chords meeting inside or outside the circle.

Tangent-secant: $PT^2 = PA \times PB$

- **P** is the external point, **T** the point of contact, **A, B** the points where the secant cuts the circle.

08 Mensuration

Right circular cylinder

Curved surface area — $2\pi rh$

Total surface area — $2\pi r(r + h)$

Volume — $\pi r^2 h$

- **Hollow cylinder** (outer R, inner r, height h):

Volume of material — $\pi h(R^2 - r^2)$

Total surface area — $2\pi h(R + r) + 2\pi(R^2 - r^2)$

Right circular cone

Slant height — $l = \sqrt{r^2 + h^2}$

Curved surface area — πrl

Total surface area — $\pi r(r + l)$

Volume — $\frac{1}{3}\pi r^2 h$

- **Cone made from a sector:** arc length of sector = circumference of cone base. Radius of sector = slant height.

Sphere & hemisphere

	Sphere	Hemisphere
Curved SA	$4\pi r^2$	$2\pi r^2$
Total SA	$4\pi r^2$	$3\pi r^2$ (incl. base)
Volume	$4/3\ \pi r^3$	$2/3\ \pi r^3$

Hollow sphere (R, r) — $4/3\ \pi(R^3 - r^3)$

Combination of solids

- **Surface area:** add only the **exposed** surfaces. Do not count the faces joined together.
- **Volume:** add the individual volumes.

Solid	Parts
Toy / tent	cone on a cylinder or hemisphere
Capsule	cylinder + 2 hemispheres
Ice-cream cone	cone + hemisphere on top
Test tube	cylinder + hemisphere

Recasting & cost problems

Volume melted = Volume of new solid(s)

Number of small solids — Volume of big ÷ Volume of one small

Volume of water flow — area of cross-section × speed × time

Cost — area (or volume) × rate

- **Conversions:** 1 m^3 = 1000 L, 1 L = 1000 cm^3, 1 m = 100 cm. Convert **all lengths to the same unit** first.

09 Trigonometry

Ratios in a right triangle

sin θ	Opposite / Hypotenuse
cos θ	Adjacent / Hypotenuse
tan θ	Opposite / Adjacent
cosec θ	1 / sin θ
sec θ	1 / cos θ
cot θ	1 / tan θ
tan θ	sin θ / cos θ
cot θ	cos θ / sin θ

Standard values

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	∞
cosec	∞	2	√2	2/√3	1
sec	1	2/√3	√2	2	∞
cot	∞	√3	1	1/√3	0

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$1 + \tan^2 A = \sec^2 A$$

$$1 + \cot^2 A = \operatorname{cosec}^2 A$$

Rearranged — $\sin^2 A = 1 - \cos^2 A$

$\sec^2 A - \tan^2 A = 1$

$\operatorname{cosec}^2 A - \cot^2 A = 1$

- **Proving identities:** work on **one side** (usually the complex side). Convert everything to sin and cos. Simplify. Do **not** cross-multiply.

Complementary angles

sin (90° − A)	cos A
cos (90° − A)	sin A
tan (90° − A)	cot A
cot (90° − A)	tan A
sec (90° − A)	cosec A
cosec (90° − A)	sec A

Heights & distances

- **Angle of elevation:** observer looks **up** from the horizontal. **Angle of depression:** looks **down** from the horizontal.

Height h, distance d — $\tan\theta = h / d$

- **Steps:** draw a neat figure → mark the right triangles → choose the ratio that uses the given and required sides → solve.

Two angles, one tower — two equations, two unknowns

√2, √3 — 1.414, 1.732

10 Statistics & probability

Mean of grouped data

- **x** = class mark = (lower + upper) / 2. **f** = frequency.

Method	Formula
Direct	$\bar{x} = \Sigma fx / \Sigma f$
Short-cut	$\bar{x} = A + \Sigma fd / \Sigma f$, $d = x - A$
Step-deviation	$\bar{x} = A + (\Sigma ft / \Sigma f) \times i$, $t = (x - A)/i$

- **A** = assumed mean (a class mark near the middle). **i** = class width.

Median, mode & ranges

Median (n odd) — **value at (n + 1)/2**

Median (n even) — **average of n/2 and (n/2 + 1)**

- **Mode:** the most frequent value. **Modal class:** the class with the highest frequency.
- **Median class:** the class containing the (n/2)th term in the cumulative table.

Range — **highest − lowest**

Histogram & ogive

- **Histogram:** bars with no gaps, class limits on the x-axis. **Mode from histogram:** in the tallest bar, join the top corners to the neighbours' corners; the x-value where the lines cross is the mode.
- **Ogive (less-than):** plot (upper limit, cumulative frequency) and join smoothly.

From ogive	Read at
Median	n / 2
Lower quartile Q_1	n / 4
Upper quartile Q_3	3n / 4

Interquartile range — $Q_3 - Q_1$

Probability

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Range — $0 \le P(E) \le 1$

Sure event — $P = 1$

Impossible event — $P = 0$

Complement — $P(\text{not } E) = 1 - P(E)$

Experiment	Sample space size
1 coin	2 (H, T)
2 coins	4 (HH, HT, TH, TT)
1 die	6
2 dice	36
Pack of cards	52

- **Cards:** 4 suits × 13. 26 red, 26 black. 12 face cards (J, Q, K). 4 aces. 13 of each suit.

11 Formula speed-run & traps

One formula per chapter

Chapter	Key formula
GST	Taxable value × rate / 100
RD	$I = P \times n(n+1)/2 \times r/1200$
Shares	Income = N × rate × FV
Quadratic	$x = (-b \pm \sqrt{D}) / 2a$
Proportion	ad = bc
Factor theorem	f(a) = 0
Matrices	AB: columns of A = rows of B
AP	$S = n/2\,[2a + (n-1)d]$
GP	$T = a\,r^{n-1}$
Section	$(m_1x_2 + m_2x_1)/(m_1 + m_2)$
Line	$y = mx + c$
Similarity	Area ratio = (side ratio)²
Circle	$PT^2 = PA \times PB$
Cone	$V = \frac{1}{3}\pi r^2 h$, $l^2 = r^2 + h^2$
Sphere	$V = 4/3\ \pi r^3$
Trig	$\sin^2 + \cos^2 = 1$
Mean	$A + (\Sigma ft/\Sigma f) \times i$
Probability	P = favourable / total

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That was the cheat sheet. Now go practise.

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Test yourself

Turn what you just revised into questions and find your weak spots.

Plan & focus

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